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Midterm Examination – March 2025

**24MAT120 Multivariable Calculus**

S2 Int. MSc. Mathematics with minor in Data Science &

S2 Int. MSc. Physics with minor in Scientific Computing

Scientific calculators are allowed

Maximum marks : 50 Time : 2 Hours

| Qn. No. | Questions | Marks | CO | Blooms taxonomy level |
|---|---|---|---|---|
| 1 | Find the limit of $f$ as $(x,y) \to (0,0)$, or show that the limit does not exist. (a) $f(x,y) = \frac{x^2y}{x^4+y^2}$ (b) $\frac{\sqrt{x}+\sqrt{y}}{x+2\sqrt{x}-y+2\sqrt{y}}$ | 6 | CO2 | L2 |
| 2 | A bug crawls on the surface $z = (x+y)^2$ directly above a path in the $xy-$plane given by $x = f(t), y = g(t)$. If $f(2) = 4, f'(2) = -1, g(2) = -2$ and $g'(2) = -3$, then at what rate is the bug's elevation $z$ changing when $t = 2$? | 6 | CO1 | L2 |
| 3 | Find a plane that is tangent to the surface $x^2+2xy-y^2+z^2 = 7$ at the point $(1,-1,3)$. | 6 | CO2 | L2 |
| 4 | Is there a direction $u$ in which the rate of change of $f(x,y) = x^2-3xy+4y^2$ at $(1,2)$ equals to 14. Justify your answer. | 6 | CO1 | L2 |
| 5 | Find the volume of the solid beneath the surface $z = 16-x^2-y^2$ and above the region $R$ bounded by the curves $y = e^x, y = 1, x = 2$ in the $xy-$plane. | 8 | CO4 | L2 |

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| 6 | Find the volume of the tetrahedron whose vertices are at (0,0,0), (3,0,0), (3,0,3) and (0,0,3). | 8 | CO4 | L2 |
| 7 | Change the Cartesian integral into an equivalent polar integral and then evaluate.<br>(a) $\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dydx$ (b) $\int_{0}^{2}\int_{0}^{\sqrt{2x-x^2}} \frac{1}{(x^2+y^2)^2} dydx$. | 10 | CO4 | L2 |

L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create